# SkillClaw: Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Closed-loop, evidence-driven skill updates from cross-user trajectories

Source: arxiv.org/abs/2604.08377

Audience: ML researchers & AI systems engineers

# Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently

**Static skill library (today)**

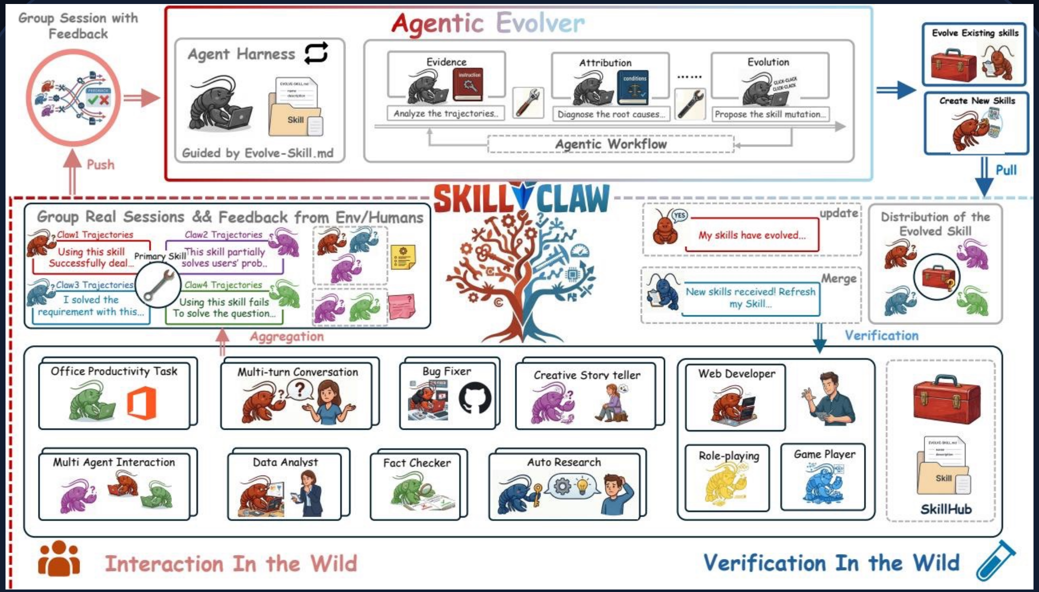
- Skills are manually installed from a hub
- Runtime fixes stay within one session
- Failures recur across users & time
- No system-level knowledge accumulation

**Collectively evolving skills (SkillClaw)**

- Aggregate trajectories across users
- Diagnose failures via evidence + attribution
- Propose skill mutations with agentic reasoning
- Verify & synchronize improvements to all agents

Why existing approaches fall short: memory stays instance-tied; skill libraries stay static.

# Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution




Centralized closed-loop: interaction → session collection → skill evolutio

## Four-stage loop (SkillClaw)

**Evidence**
Analyze skill-grouped trajectories

**Attribution**
Diagnose root causes of failures/success

**Evolution**
Propose skill mutations (refine/create)

**Distribution**
Verify, merge, and synchronize to agents

Evidence from many users closes the loop on shared skills.

# Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

### 1) Evidence
Analyze trajectories to identify behavioral patterns

### 2) Attribution
Diagnose root causes of failures and successes

### 3) Evolution
Propose skill mutations (refine existing / create new)

Key point: evolution is open-ended agent reasoning over evidence — not predefined update rules.

# Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Agents emit structured session trajectories with full action–feedback causal chains

Trajectories are aggregated across users and grouped by referenced skills

Diverse contexts reveal complementary signals: success patterns + failure modes

One user rarely yields enough evidence to separate generalizable improvements from idiosyncratic fixes

**Compatible systems**

- OpenClaw
- CoPaw
- IronClaw
- PicoClaw
- ZeroClaw
- NanoClaw
- NemoClaw

# Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

| Date | Outcome | Rationale (from evolution log) |
| --- | --- | --- |
| 2026-04-01 | Accepted | validate-tmp-workspace-inputs; upgrade best pool |
| 2026-04-02 | Rejected | multimodal input validation & env init; keep pool |
| 2026-04-03 | Rejected | creative pipeline extract paths; keep pool |
| 2026-04-04 | Rejected | added classification + validation; keep pool |
| 2026-04-05 | Rejected | fail-fast per-file validation; keep pool |
| 2026-04-06 | Rejected | extended PDF-to-poster/document-to-visual; not admitted |

**Validator accepted 1/6 candidates (conservative verification)**

# Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

| Date | Outcome | Rationale (from evolution log) |
|---|---|---|
| 2026-04-01 | Accepted | safe fallback vs blocking on push failure |
| 2026-04-02 | Accepted | unified patch/bundle filenames; reduced inconsistency |
| 2026-04-03 | Accepted | auth alternatives + secrets audit; fixed subshell pitfalls |
| 2026-04-04 | Accepted | same-pool retest; validator confirmed best |
| 2026-04-05 | Rejected | treat push hang as auth failure; no improvement |
| 2026-04-06 | Rejected | identity config + filename consistency; no improvement |

Validator accepted 3/6 updates (then plateau)

# Key Takeaways: Three Properties That Distinguish SkillClaw from Existing Systems

Collective evolution: individual interactions improve a shared ecosystem

Fully automatic: runtime-driven updates without manual curation

Agentic evolution paradigm: open-ended reasoning over evidence (not fixed rules)

Implication: design agent ecosystems where deployed skills continuously improve from cross-user evidence, while verification guards against regressions.